

Parallel Robots

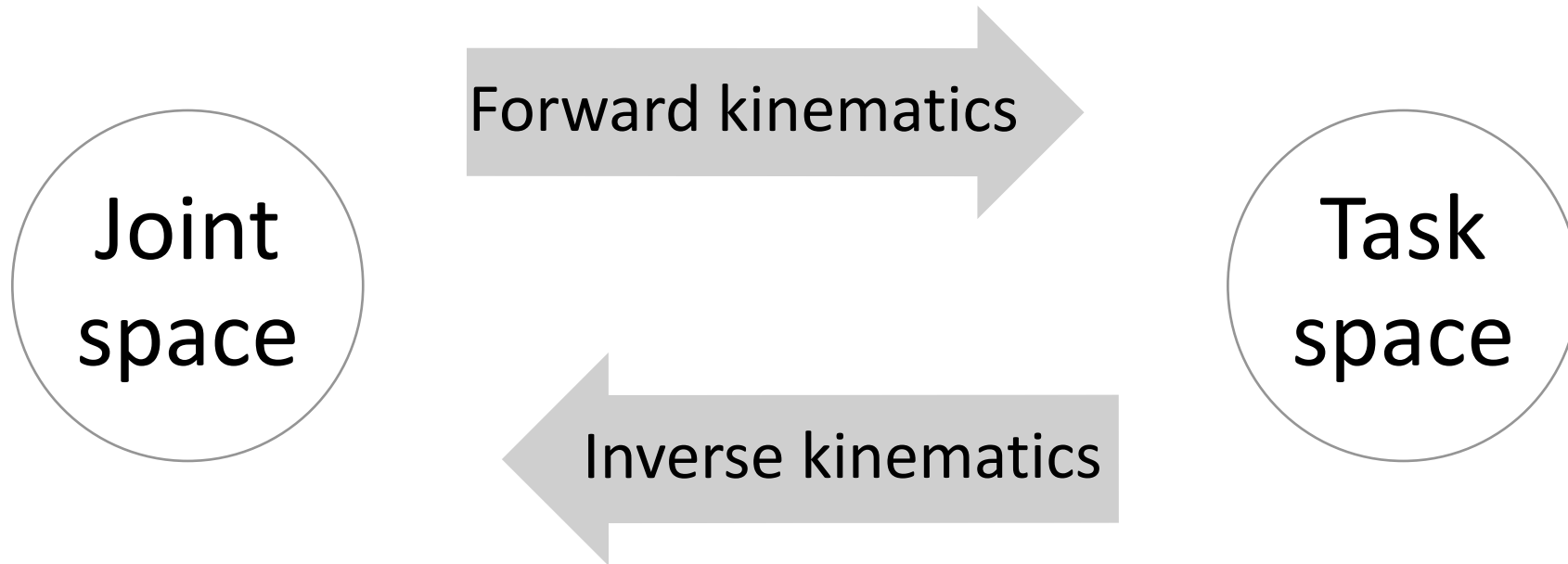
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Forward and Inverse Kinematics

- Goal: establishing the relations between joint-space coordinates and task-space coordinates



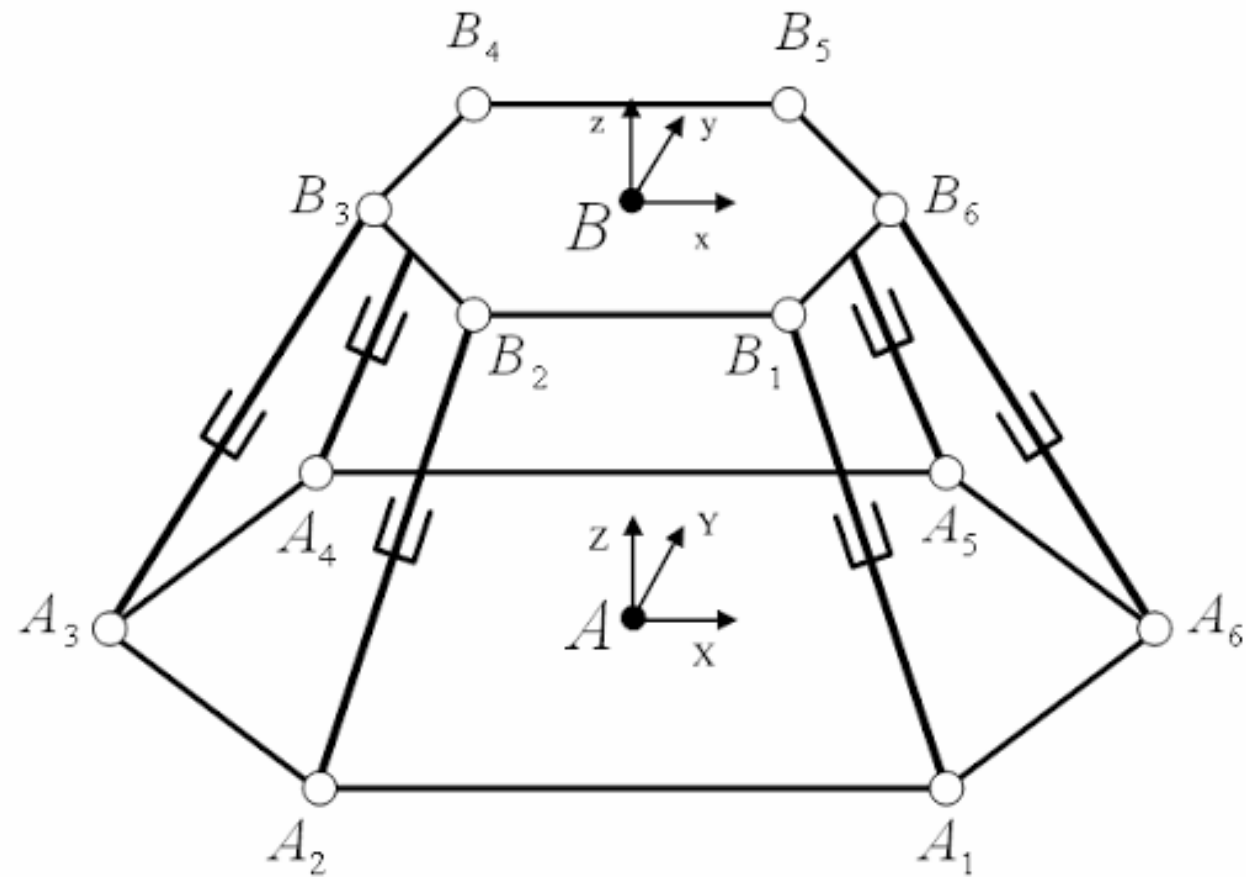
- 2-RPR robot
 - How to actuate?
 - How to parametrize? $\rightarrow$ joint-space and task-space coordinates
 - How many assembly modes?
 - How many working modes?

- Joint-space coordinates: $\mathbf{q} \in \mathbb{R}^n$ or $\mathbf{q}_a \in \mathbb{R}^n$
 - Actuated joint coordinates, describing an angle or a signed length
- Passive joint coordinates: $\mathbf{q}_p \in \mathbb{R}^{n_p}$
- Task-space coordinates: depends on the robot and on the task
 - Generalized coordinates: $\mathbf{x} \in \mathbb{R}^m$ with $m \geq n$ (why?)
 - Position, Euler angles, quaternions, rotation matrices...
 - Homogeneous transformation: $\mathbf{T} \in SE(3)$
 - How many independent coordinates? How many constraint equations?
 - What about redundant robots?

- Main goal of geometric modeling
 - Defining the robot configuration from a given set of coordinates
- Common geometric models
 - Implicit: $f(\mathbf{q}, \mathbf{x}) = 0$
 - Inverse (IK): $\mathbf{q} = f_{IK}(\mathbf{x})$
 - Forward (FK): $\mathbf{x} = f_{FK}(\mathbf{q})$
- Main approach: loop-closure equations

- Goal: finding joint-space coordinates from task-space coordinates: $\mathbf{q} = \mathbf{f}_{IK}(\mathbf{x})$
- Why?
- On parallel robots, IK are usually simple $\rightarrow$ closed-form solution
 - Why?
- Working modes = solutions of the IK
- How to choose the working mode?

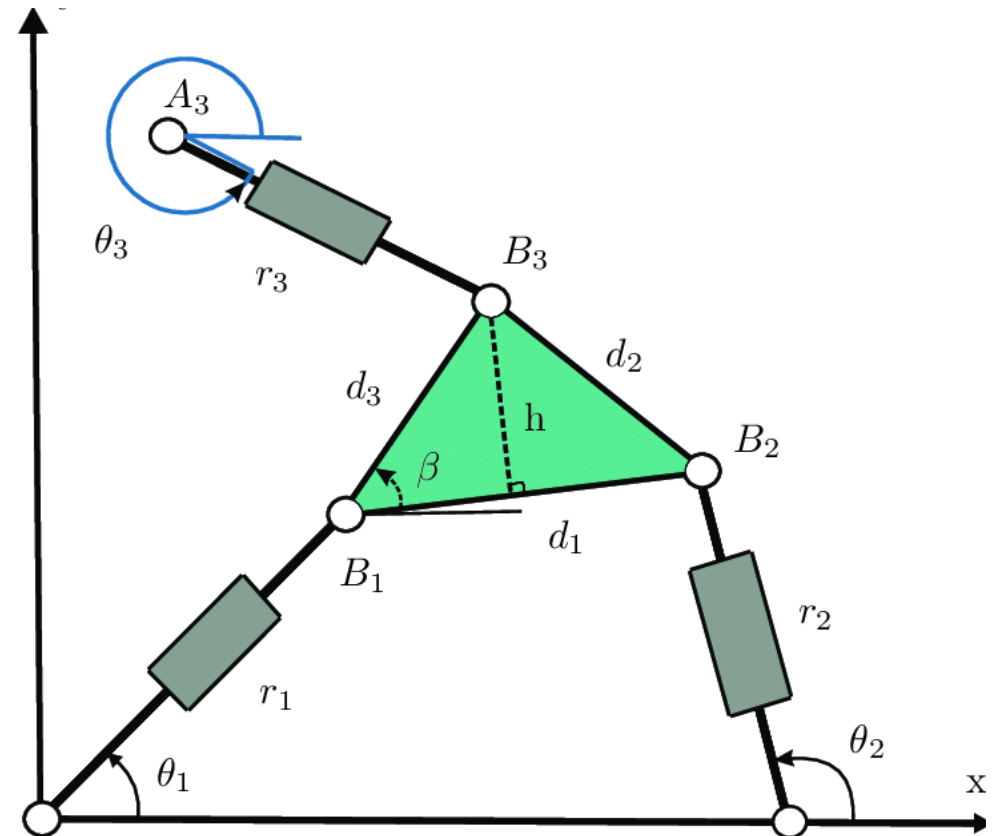
- Example: Gough-Stewart platform



<https://doi.org/10.1115/IMECE2009-11148>

- Goal: finding task-space coordinates from joint-space coordinates: $\mathbf{x} = \mathbf{f}_{FK}(\mathbf{q})$
- Why?
- On parallel robots, FK usually have no closed-form solution
 - What to do?
- Assembly modes = solutions of the FK
- How to choose the assembly mode?

- Example: 5-bar mechanism
 - Analytical solutions
- Example: 3-RPR robot
 - Analytical solutions → exercise
- Example: Gough-Stewart platform
 - No known analytical solution
 - Up to 40 distinct solutions



<https://doi.org/10.1016/j.mechmachtheory.2018.05.004>

- There are several solutions to FK
- How to choose the assembly mode corresponding to the actual platform pose?
 - Sensor redundancy
 - Exteroception
 - Numerical methods with theoretical guarantees
 - E.g., interval methods, Kantorovitch's theorem

- Newton's methods (or Newton-Raphson method)

Differential Kinematics and Jacobian Matrices

Method 1 – Differentiation

- Goal: establishing the relations between joint-space velocity and task-space velocity
- Some notations:
 - $\dot{\mathbf{q}}_a$ or $\dot{\mathbf{q}}$: active joint velocities = time-derivative of joint-space coordinates
 - $\mathbf{t}_p \in \mathfrak{so}(3)$: end-effector twist (how to represent?)
 - $\dot{\mathbf{x}}$: time-derivative of task-space coordinates
- For a spatial robot, often $\mathbf{t}_p \neq \dot{\mathbf{x}}$. Why?

- We can define Jacobian matrices of a parallel robot as the partial derivatives of the implicit kinematics $\mathbf{f}(\mathbf{x}, \mathbf{q}) = \mathbf{0}$

- Differentiation w.r.t. time:

$$\mathbf{A}(\mathbf{x}, \mathbf{q})\dot{\mathbf{x}} + \mathbf{B}(\mathbf{x}, \mathbf{q})\dot{\mathbf{q}} = \mathbf{0}$$

- With $\mathbf{A} = \frac{\partial \mathbf{f}}{\partial \mathbf{x}}$ and $\mathbf{B} = \frac{\partial \mathbf{f}}{\partial \mathbf{q}}$ the Jacobian matrices
- Matrices $\mathbf{A}$ and $\mathbf{B}$ are not unique. Why?
- What are the dimensions? Consider non-redundant robots

- Goal: establishing the relations between joint-space acceleration and task-space acceleration

- Differentiation w.r.t. time:

$$A\ddot{x} + \dot{A}\dot{x} + \dot{B}\dot{q} + B\ddot{q} = 0$$

- Then

$$\begin{aligned}\ddot{x} &= -A^{-1}(\dot{A}\dot{x} + \dot{B}\dot{q} + B\ddot{q}) \\ \ddot{q} &= -B^{-1}(A\ddot{x} + \dot{A}\dot{x} + \dot{B}\dot{q})\end{aligned}$$

Differential Kinematics and Jacobian Matrices

Method 2 – Geometric Approach and Screw Theory
(Following Briot's and Khalil's approach)

- Consider that “parallel robot = multiple serial legs connected together”
- Then, we can use the Jacobian matrices of each leg
- For the i-th serial leg, $\mathbf{t}_p = \mathbf{J}_i(\mathbf{q}_i)\dot{\mathbf{q}}_i$ where $\mathbf{J}_i = [\mathbf{J}_{i1} \quad \dots \quad \mathbf{J}_{im_i}]$
- Thus,

$$\mathbf{t}_p = \sum_j \mathbf{J}_{ij} \dot{q}_{ij}$$

- $\mathbf{J}_{ij}$ is the unit twist associated to the j-th joint at the i-th leg

$$\mathbf{J}_{ij} = \begin{bmatrix} \sigma_{ij} \mathbf{a}_{ij} + \bar{\sigma}_{ij} (\mathbf{a}_{ij} \times \mathbf{r}_{A_{ij}P}) \\ \bar{\sigma}_{ij} \mathbf{a}_{ij} \end{bmatrix}$$

$\mathbf{a}_{ij}$ is the joint axis

$\sigma_{ij} = 1 - \bar{\sigma}_{ij} = 1$ if P joint

$\sigma_{ij} = 1 - \bar{\sigma}_{ij} = 0$ if R joint

- Let $\$$ be a unit twist and ζ a unit wrench. Then, $\$$ and ζ are reciprocal if

$$\zeta^T \$ = 0$$

- What is the physical meaning of this reciprocal product?
- Now, let us separate active and passive joints

$$\mathbf{t}_p = [\$_{ia} \quad \$_{ip}] \begin{bmatrix} \dot{\mathbf{q}}_{ai} \\ \dot{\mathbf{q}}_{pi} \end{bmatrix}$$

- Goal: eliminate passive joints

- To eliminate passive joint velocities: find reciprocal wrench ζ_i such that

$$\zeta_i^T \$_{ip} = \mathbf{0}$$

and $\zeta_i^T \$_{ip}$ has no zero components. Then,

$$\zeta_i^T \mathbf{t}_p = \zeta_i^T \$_{ia} \dot{q}_{ai}$$

- There are as many independent wrenches ζ_i as active joints
- Combining all legs,

$$\begin{bmatrix} \zeta_1^T \\ \zeta_2^T \\ \dots \\ \zeta_n^T \end{bmatrix} \mathbf{t}_p = \begin{bmatrix} \zeta_1^T \$_{1a} & \mathbf{0} & \dots & \mathbf{0} \\ \mathbf{0} & \zeta_2^T \$_{2a} & \dots & \mathbf{0} \\ \dots & \dots & \dots & \dots \\ \mathbf{0} & \mathbf{0} & \dots & \zeta_n^T \$_{na} \end{bmatrix} \dot{\mathbf{q}}$$

- Let us consider a matrix $\mathbf{C} = \mathbf{C}(\mathbf{x})$ such that

$$\mathbf{t}_p = \mathbf{C}\dot{\mathbf{x}}$$

- Then,

$$\mathbf{A}\dot{\mathbf{x}} + \mathbf{B}\dot{\mathbf{q}} = \mathbf{0}$$

where

$$\mathbf{A} = \mathbf{A}_x \mathbf{C}$$

$$\mathbf{A}_x = [\boldsymbol{\zeta}_1 \quad \dots \quad \boldsymbol{\zeta}_n]^T$$

$$\mathbf{B} = \begin{bmatrix} \boldsymbol{\zeta}_1^T \mathbf{\$}_{1a} & \mathbf{0} & \dots & \mathbf{0} \\ \mathbf{0} & \boldsymbol{\zeta}_2^T \mathbf{\$}_{2a} & \dots & \mathbf{0} \\ \dots & \dots & \dots & \dots \\ \mathbf{0} & \mathbf{0} & \dots & \boldsymbol{\zeta}_n^T \mathbf{\$}_{na} \end{bmatrix}$$

- How to find $\mathcal{C}$ considering roll-pitch-yaw angles to represent orientations?

- Find the reciprocal wrenches for 3-RRR and 3-RPR robots

Statics

- Consider robot kinematics

$$\dot{\mathbf{q}} = \mathbf{J}^{-1} \dot{\mathbf{x}}$$

- Using virtual work principle, we can show that

$$\mathbf{J}^{-T} \boldsymbol{\tau} = \mathbf{w}$$

with $\boldsymbol{\tau}$ generalized actuation forces and $\mathbf{w}$ the output wrench

- For robots that are **not underactuated**, the statics only depend on the geometry
- Statics model is useful for simplified dynamics models
- Example: only consider inertia at the actuators and/or end-effector

- Planar robot pushing towards the wall

Singularity Analysis

- Consider a parallel robot such that

$$\mathbf{A}(\mathbf{x}, \mathbf{q})\dot{\mathbf{x}} + \mathbf{B}(\mathbf{x}, \mathbf{q})\dot{\mathbf{q}} = \mathbf{0}$$

where $\mathbf{A}$ and $\mathbf{B}$ are square matrices

- Definition: a configuration is singular if either $\mathbf{A}$ or $\mathbf{B}$ is singular
- With this definition, a configuration is singular for a given set of $\mathbf{q}$ and $\mathbf{x}$

- This is the historical definition used by Gosselin (1990)
- More recently, more complete definitions were proposed, but are out of scope
- One of the most important and difficult topic for the study of parallel robot
- Parallel robots exhibit different singularities compared to serial architectures

- Definition: the first kind of singularity occurs when

$$\det \mathbf{B} = 0$$

- Equivalently: there exist $\dot{\mathbf{q}} \neq \mathbf{0}$ such that $\dot{\mathbf{x}} = \mathbf{0}$, so $\dot{\mathbf{q}} \in \ker \mathbf{B}$
- Same as the singularity of a serial robot

- Different branches (solutions) of the IK meet
- Physical interpretation
 - Workspace boundary
 - The robot loses the ability to move along one direction
 - The actuators require infinite force to move along one direction

- Definition: the second kind of singularity occurs when

$$\det \mathbf{A} = 0$$

- Equivalently: there exist $\dot{\mathbf{x}} \neq \mathbf{0}$ such that $\dot{\mathbf{q}} = \mathbf{0}$, so $\dot{\mathbf{x}} \in \ker \mathbf{A}$
- Only for parallel robots
- Happens within the workspace

- Different branches (solutions) of the FK meet
- Physical interpretation
 - The end-effector can move even when all actuators are locked
 - The actuators require infinite force to resist to some disturbances

- Identify serial and parallel singularities of 5-bar and 3-RRR robots
 - Geometrical interpretation

Serial singularities

- Most of the time, it is enough to analyze $\det \mathbf{B}$
- Often, $\mathbf{B}$ is block diagonal
- Can be geometrically “intuitive” as it describes the boundaries of the workspace
- Often, it is sufficient to analyze the leg mechanism

Parallel singularities

- In most cases, there are no closed-form solutions to $\det \mathbf{A} = 0$
- Geometric interpretation in necessary
- Numerical solutions do not help geometric understanding

- Let us revisit the differential kinematics obtained with screw theory approach

$$\begin{bmatrix} \zeta_1^T \\ \zeta_2^T \\ \dots \\ \zeta_n^T \end{bmatrix} \mathbf{t}_p = \begin{bmatrix} \zeta_1^T \$_{1a} & \mathbf{0} & \dots & \mathbf{0} \\ \mathbf{0} & \zeta_2^T \$_{2a} & \dots & \mathbf{0} \\ \dots & \dots & \dots & \dots \\ \mathbf{0} & \mathbf{0} & \dots & \zeta_n^T \$_{na} \end{bmatrix} \dot{\mathbf{q}}$$

- Under some assumptions... (which ones?)
- Serial singularity if $\zeta_i^T \$_{ia} = 0$
 - The screw that is reciprocal to passive joints is also reciprocal to the active one(s)
- Parallel singularity if there is a linear dependency between $(\zeta_i)_i$

- Serial singularity example – $\underline{R}RR$ and $R\underline{P}R$ legs

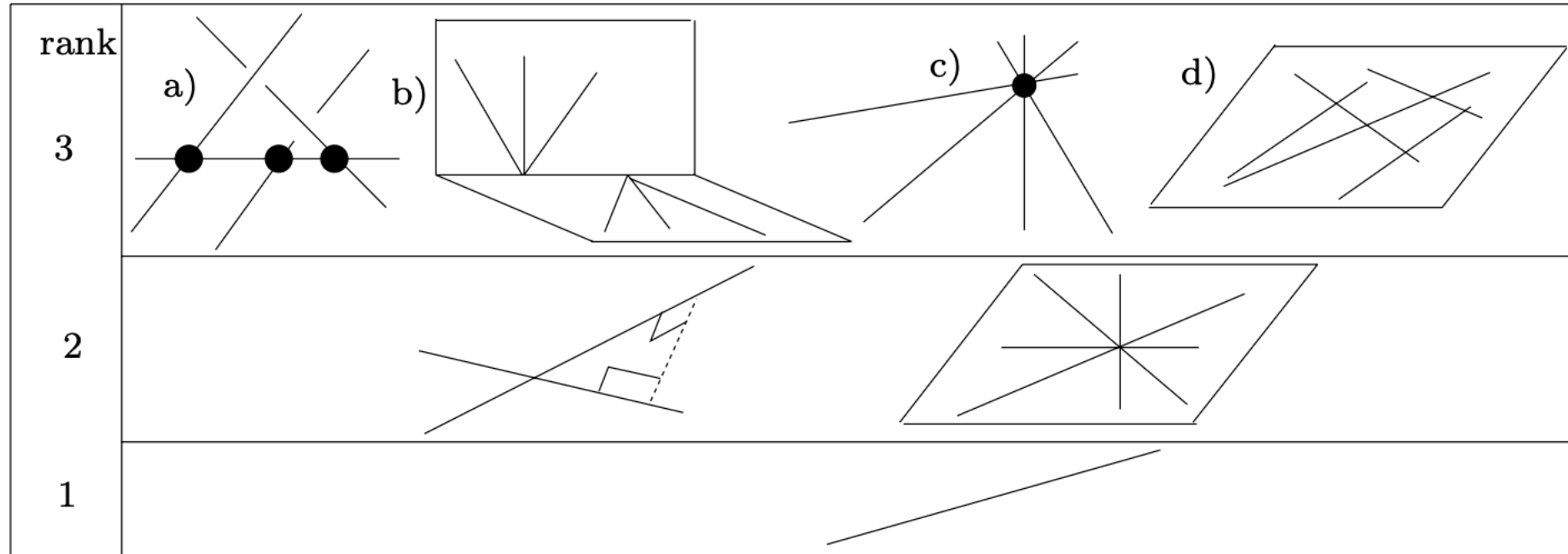
- Parallel singularity if there is a linear dependency between $(\zeta_i)_i$
- How to detect geometrically a linear dependency?
 - Grassmann geometry
 - Grassman-Cayley algebra

- Definition: the Plücker coordinates of a line of direction $\mathbf{f}$ passing through a point P are

$$\begin{bmatrix} \mathbf{f} \\ \mathbf{r}_{OP} \times \mathbf{f} \end{bmatrix}$$

- Theorem: for any spatial (planar) robot whose legs are attached to the end-effector through a spherical (revolute) joint, $\mathbf{A}$ matrix will involve the Plücker vector of a line associated to the link connecting the leg to the end-effector
- In layman's terms, ζ_i is a Plücker vector

- Grassmann geometry allows for evaluating the rank of the A matrix



<https://doi.org/10.1007/1-4020-4133-0>

- Grassmann geometry allows for evaluating the rank of the A matrix
- For 2-DOF planar robots, rank deficiency if all two lines are coincident
- For 3-DOF planar robots, rank deficiency if all three lines are concurrent
 - Parallel lines are concurrent at infinity
- 3D case is out-of-scope